Andhra University College of Engineering

Information Technology | Course Code: IT301

Web Technologies & Architecture - Full-Stack Web Architecture & REST API De

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Overview: DOM traversal, event loop mechanisms, JWT authentication, relational ORM data mapping

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Semantic Markup & Modern Styling
HTML5 semantic elements, CSS Flexbox and Grid specifications, Media queries, Mobile-first responsive

Unit 2: JavaScript Runtime & Asynchronous Engine
Prototypal inheritance, Closures, Promises, Async/Await syntax, Event loop, DOM API manipulations

Unit 3: Server Middleware & RESTful API Architecture
HTTP/1.1 vs HTTP/2 protocol, REST verbs, JSON payloads, Routing, CORS policies, JWT token authentica

Unit 4: Web Application Security & Storage
XSS, CSRF, SQL injection defenses, Content Security Policy, LocalStorage vs Cookies, Service Workers

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Explain the JavaScript Event Loop, Call Stack, Task Queue, and Microtask Queue with a timing execution
2. Design a scalable RESTful API specification for an institutional course enrollment system adhering to
3. Describe Cross-Site Scripting XSS attack vectors and present automated CSP mitigation configurations.

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* David Flanagan - JavaScript: The Definitive Guide O Reilly 7th Ed
* Kyle Simpson - You Don t Know JS Yet Series

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